Article

Crossover or First-Order: That is the Question

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Abstract

The equation of state of hot and dense nuclear matter plays a fundamental role in many areas. However, owing to the ~~non-perturbative~~ nature of strong interaction, a reliable treatment is still in debate. We use a ~~contact interaction~~ to study the related issues at finite temperature and quark chemical potential, and carefully compare a ~~non-covariant~~ and a covariant regularization scheme. The numerical results show that the ~~nature~~ of the phase transition depends sensitively on the regularization scheme and parameter choices.

Keywords: crossover; first-order phase transition; equation of state; finite chemical potential; contact interaction; Dyson-Schwinger equations; proper-time regularization

1. Introduction

The Standard Model of particle physics stands as the most successful theoretical framework in modern high-energy physics, having been subjected to and passed a wide range of experimental tests over the past several decades. It provides a unified description of three of the four known fundamental interactions in nature: electromagnetic, weak, and strong interactions. Among these, the strong interaction, which is described by the theory of quantum chromodynamics (QCD), is of particular importance. QCD is a non-Abelian gauge theory based on the $SU(3)_c$ color group, where quarks interact via the exchange of gluons ~~[1,2]~~. While perturbation theory has been proven to be extremely powerful at high energies, where the coupling constant becomes weak ~~due~~ to asymptotic freedom, the situation changes dramatically at low energies. In the infrared regime, the QCD coupling grows large ~~[3–8]~~ and renders perturbative methods ineffective, thereby necessitating alternative non-perturbative approaches.

The thermodynamics of strongly interacting matter, encapsulated in the equation of state (EoS) of QCD, plays a fundamental role in both cosmology and astrophysics. In cosmology, the QCD EoS governs the dynamics of the early universe during the quark-hadron transition, possibly influencing the evolution of primordial fluctuations and thus potentially leaving observable imprints in the cosmic microwave background or relic abundances [9–11]. In astrophysics, it dictates the internal structure and stability of compact stellar objects, such as neutron stars and hybrid stars. Accurate knowledge of the EoS at high baryon density, or chemical potential (μ), is essential for determining properties such as maximum stellar masses, radii, and tidal deformabilities, which can now be constrained by multimessenger observations, including gravitational-wave detections from binary neutron star mergers [12,13]. Thus, a quantitative and reliable determination of the QCD phase structure and its corresponding EoS is a central challenge of modern theoretical physics.

Despite extensive progress achieved through lattice QCD simulations, serious limitations remain. At vanishing chemical potential, lattice QCD has provided quantitatively reliable results for the EoS, the pseudo-critical temperature of the crossover transition, and

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fluctuations of conserved charges. However, at finite μ, lattice methods are plagued by the ~~notorious sign problem, arising from the complex phase of the fermion determinant~~ [14]. This makes straightforward Monte Carlo sampling impractical and severely restricts applicability in the low-temperature T, high-μ domain, precisely the regime of relevance for neutron star interiors and heavy-ion collisions at lower beam energies. While methods such as reweighting, Taylor expansions, or analytic continuation from imaginary chemical potential have been explored, none of them yield a fully controlled and systematically improvable approach at large μ [15,16].

In light of these obstacles, a variety of effective models have been proposed to capture essential features of QCD in regimes inaccessible to lattice simulations. Among them, for example, the Nambu-Jona-Lasinio (NJL) model and its Polyakov-loop extended version (PNJL) have been widely employed to investigate chiral symmetry breaking, its restoration at finite T and μ, as well as qualitative aspects of the phase diagram. The NJL model successfully incorporates spontaneous chiral symmetry breaking and dynamical mass generation, which are hallmarks of low-energy QCD [17–32]. Nevertheless, it also has well-known shortcomings: the absence of confinement, dependence on ultraviolet cutoff schemes that violate Poincaré covariance, and a strong sensitivity of predictions to parameter choices and regularization prescriptions [33,34]. As a result, while the NJL framework provides valuable intuition, its quantitative reliability remains questionable.

On the other hand, another popular continuum, non-perturbative approach that preserves fundamental symmetries is the Dyson-Schwinger equations (DSEs), which offer a system of coupled integral equations for Green's functions that are valid across all energy scales [35–38]. In particular, DSEs respect Poincaré covariance and allow for the simultaneous study of confinement, dynamical chiral symmetry breaking, and their interrelation at zero or finite T and μ. While truncations of the infinite tower of equations are unavoidable, systematic schemes such as the rainbow-ladder approximation and beyond have been proven capable of producing results consistent with both lattice QCD and experimental observations in many contexts; for example, Refs. [39–49].

In this work, we adopt a rainbow-truncated quark DSE with a local contact kernel to explore features of the QCD phase structure at finite (T, μ). In this set of approximations, the bulk thermodynamics is practically mean-field equivalent to NJL-type models [43]. Crucially, because the contact interaction is nonrenormalizable, the regulator is part of the model. As we demonstrate, this leads to situations where the choice of the regularization scheme—not the interaction—drastically affects the behavior of matter under extreme conditions, and can determine whether the phase transition at low T is a crossover or first-order.

2. Framework

Quark DSE and order parameters

At finite temperature and chemical potential -(T, μ), the quark gap equation, or DSE for the dressed-quark propagator, can be written as (in Euclidean space) [35–38]

$$\begin{aligned} S^{-1}(\vec{p},\tilde{\omega}_n) &= i\vec{\gamma}\cdot\vec{p} + i\gamma_4\tilde{\omega}_n + m \\ &\quad + \int_{l,q} g^2 D_{\mu\nu}(\vec{p}-\vec{q},\Omega_{nl};T,\mu)\gamma_\mu \frac{\lambda^a}{2} S(\vec{q},\tilde{\omega}_l)\Gamma^a_\nu(\vec{p},\tilde{\omega}_n,\vec{q},\tilde{\omega}_l;T,\mu), \end{aligned} \tag{1}$$

where $\tilde{\omega}_n = \omega_n + i\mu$, $\omega_n = (2n+1)\pi T$, with $n \in \mathbb{Z}$ the fermion Matsubara frequencies, m is the current quark mass, $\int_{l,q} := T\sum_{l=-\infty}^{\infty}\int_{-\infty}^{\infty}\frac{d^3q}{(2\pi)^3}$, $\Omega_{nl} = \tilde{\omega}_n - \tilde{\omega}_l = 2(n-l)\pi T$, and the dressed-quark propagator has the general form

$$S^{-1}(\vec{p},\tilde{\omega}_n) = i\vec{\gamma}\cdot\vec{p}A(\vec{p},\tilde{\omega}_n) + i\gamma_4\tilde{\omega}_n C(\vec{p},\tilde{\omega}_n) + B(\vec{p},\tilde{\omega}_n). \tag{2}$$

The scalar functions, $F = A, B, C$, are complex and satisfy

$$F(\vec{p}, \tilde{\omega}_n)^* = F(\vec{p}, \tilde{\omega}_{-n-1}). \tag{3}$$

For the gluon propagator, a particularly simple and useful model is ~~the contact~~ (momentum-independent) vector $\otimes$ vector quark-quark interaction [50],

$$g^2 D_{\mu\nu} = \delta_{\mu\nu} \frac{4\pi\alpha_{ir}}{m_G^2}, \tag{4}$$

where α_{ir} is the zero-momentum value of the running-coupling constant ~~[3–8]~~, and m_G is a gluon mass-scale that generated dynamically in QCD.

We can see that, ~~just like the NJL model~~, this ~~form~~ doesn't have momentum dependence, so that related equations are largely algebraic, derivations and formulae ~~would remain~~ highly transparent, which makes the outcomes easy to interpret and even enables meaningful comparisons with results from more sophisticated ~~frameworks~~. The physical relevance of the ~~results of contact interaction~~ can also be appreciated under careful interpretation. Therefore, it is widely used in many topics; for example, Refs. [51–59].

With these treatments, and taking the "Rainbow approximation", $\Gamma_\nu^a(\vec{p}, \tilde{\omega}_n, \vec{q}, \tilde{\omega}_l; T, \mu) = \frac{\lambda^a}{2}\gamma_\nu$, for the dressed quark-gluon vertex, the scalar functions ~~would then~~ satisfy,

$$\begin{aligned}
&A(\vec{p}, \tilde{\omega}_n) = 1, \\
&\tilde{\omega}_n C(\vec{p}, \tilde{\omega}_n) = \tilde{\omega}_n + \frac{32\pi}{3} \frac{\alpha_{ir}}{m_G^2} \int_{l,q} \frac{\tilde{\omega}_l C(\vec{q}, \tilde{\omega}_l)}{\vec{q}^2 + \tilde{\omega}_l^2 C^2(\vec{q}, \tilde{\omega}_l) + B^2(\vec{q}, \tilde{\omega}_l)}, \\
&B(\vec{p}, \tilde{\omega}_n) = m + \frac{64\pi}{3} \frac{\alpha_{ir}}{m_G^2} \int_{l,q} \frac{B(\vec{q}, \tilde{\omega}_l)}{\vec{q}^2 + \tilde{\omega}_l^2 C^2(\vec{q}, \tilde{\omega}_l) + B^2(\vec{q}, \tilde{\omega}_l)}.
\end{aligned} \tag{5}$$

Since the ~~integral~~ here is ~~momentum~~ $\vec{p}$ independent, while the sum is $\tilde{\omega}_n$ independent, a solution at one value of $\vec{p}$ and $\tilde{\omega}_n$ is the solution at all values; viz., any nonzero solution must be of the form,

$$\begin{aligned}
&B(\vec{p}, \tilde{\omega}_n) \equiv R_B, \\
&\tilde{\omega}_n (C(\vec{p}, \tilde{\omega}_n) - 1) \equiv -iR_C,
\end{aligned} \tag{6}$$

and Equation 3 entails that R_B and R_C are real numbers. Then,

$$\begin{aligned}
R_C &= \frac{8}{3\pi} \frac{\alpha_{ir}}{m_G^2} \int_{n,s} s^{\frac{1}{2}} \frac{i\tilde{\omega}'_l}{s + \tilde{\omega}'^2_n + R_B^2}, \\
R_B &= m + \frac{16}{3\pi} \frac{\alpha_{ir}}{m_G^2} \int_{n,s} s^{\frac{1}{2}} \frac{R_B}{s + \tilde{\omega}'^2_n + R_B^2},
\end{aligned} \tag{7}$$

where $\mu' = \mu - R_C$, $\tilde{\omega}'_n = \omega_n + i\mu'$; $s = \vec{q}^2$ and $\int_{n,s} := T\sum_{n=-\infty}^{\infty} \int_0^\infty ds$. We can see that mathematically R_C is always linked with μ, and plays a role in reducing it. Sometimes μ' is also referred to as the renormalized chemical potential ~~in literature~~ [21].

Equation 7 could also be written in other forms, defining $E_{s,R_B} = \sqrt{s + R_B^2}$, then

$$\begin{aligned}
R_C =& \frac{4}{3\pi} \frac{\alpha_{ir}}{m_G^2} \int_s s^{\frac{1}{2}} [n_+ - n_-], \\
R_B =& m + \frac{8}{3\pi} \frac{\alpha_{ir}}{m_G^2} \int_s \frac{R_B s^{\frac{1}{2}}}{E_{s,R_B}} [n_+ + n_-],
\end{aligned} \tag{8}$$

where

$$\begin{aligned} n_{+} &= T\sum_{n=-\infty}^{\infty}\frac{1}{E_{s,R_B}-i\tilde{\omega}'_n} = T\sum_{n=-\infty}^{\infty}\frac{E_{s,R_B}+\mu'}{(E_{s,R_B}+\mu')^2+\omega_n^2}, \\ n_{-} &= T\sum_{n=-\infty}^{\infty}\frac{1}{E_{s,R_B}+i\tilde{\omega}'_n} = T\sum_{n=-\infty}^{\infty}\frac{E_{s,R_B}-\mu'}{(E_{s,R_B}-\mu')^2+\omega_n^2}. \end{aligned} \tag{9}$$

To provide a quantitative way to distinguish different phases, ~~usually order parameters are~~ defined in phase transition studies. The quark condensate (scalar density) is one order parameter for the chiral symmetry restoration transition, defined via the dressed-quark propagator ~~[27,60]~~,

$$\langle\bar{\psi}\psi\rangle = -N_C\int_{n,q} tr_D S(\vec{q},\tilde{\omega}_n) = -N_C\frac{3}{16\pi}\frac{m_G^2}{\alpha_{ir}}(R_B-m), \tag{10}$$

where $N_C = 3$ is the quark color number. On the other hand, the total quark number density (vector density) is another popular order parameter ~~[17,21]~~, especially for finite μ areas,

$$\langle\psi^\dagger\psi\rangle = \langle\bar{\psi}\gamma_4\psi\rangle = -N_C\int_{n,q} tr_D\gamma_4 S(\vec{q},\tilde{\omega}_n) = N_C\frac{3}{8\pi}\frac{m_G^2}{\alpha_{ir}}R_C. \tag{11}$$

From Equations 10 and 11, we can clearly see the physical meaning of R_B and R_C. R_B is usually called the effective quark mass, while R_C is closely related to quark number; hence, it is not surprising that R_C has the opposite effect to that of μ.

To ~~clarify~~ the similarities and differences with the formulation of the NJL model, we present the two sets of equations side by side for comparison.

In the NJL model, the corresponding equations before regularization are

$$\begin{aligned} M &= m + G_{NJL}\frac{N_C N_f}{\pi^2}M\int_s \frac{s^{\frac{1}{2}}}{E_{s,M}}[n_+ + n_-], \\ \mu' &= \mu - g_{NJL}\frac{N_f}{2\pi^2}\int_s s^{\frac{1}{2}}[n_+ - n_-], \end{aligned} \tag{12}$$

where M is the constituent mass, g_{NJL} is the effective coupling constant of the four-fermion interaction, and

$$G_{NJL} = \frac{4N_C+1}{4N_C}g_{NJL} \tag{13}$$

is the renormalized coupling constant [21].

Comparing ~~Equation 8~~ and Equation 12, they exhibit a strong resemblance to each other. It may be expected that, with a proper definition of the corresponding coefficients, the two equations would coincide exactly. That is, by defining

$$\begin{aligned} G_{CI} &:= \frac{8\pi}{3N_C N_f}\frac{\alpha_{ir}}{m_G^2}, \\ g_{CI} &:= \frac{8\pi}{3N_f}\frac{\alpha_{ir}}{m_G^2}, \end{aligned} \tag{14}$$

the two equations will assume exactly the same form. However, that means

$$G_{CI} = \frac{1}{N_C}g_{CI}, \tag{15}$$

which is different from the expression of the NJL model Equation 13, indicating a mismatch in the corresponding coefficients.

2.2. Regularization schemes

Obviously, ~~just as in~~ the NJL model, ~~this contact interaction~~ is also non-renormalizable; therefore, a regularization scheme is unavoidable ~~to make the physical results finite and meaningful~~. ~~However,~~ this regularization should maximally preserve the properties of the model and symmetry ~~[17],~~ Currently, several schemes are available, each in some sense determines the model. For example, two of the most popular regularization schemes are the noncovariant but physically intuitive three-momentum (3M) cutoff regularization [17,19,21,27,29,33,34] and a covariant regularization in proper time (PT) form [17,20,22,23,31,32,39,43,50–59]. ~~Here,~~ we will compare them carefully.

2.2.1. Three-momentum cutoff regularization

This regularization doesn't take into account the contributions from momenta larger than a cutoff Λ_{3M}, i.e., integrates only within $s < \Lambda_{3M}^2$. Namely, the gap equation in Equation 7 becomes

$$R_B = m + \frac{16}{3\pi}\frac{\alpha_{ir}}{m_G^2}\int_{n,s}^{\Lambda_{3M}^2} s^{\frac{1}{2}}\frac{R_B}{s+\widetilde{\omega}_n'^2+R_B^2}. \tag{16}$$

2.2.2. Proper time regularization

The key treatment of this regularization is

$$\frac{1}{\mathcal{T}} = \int_0^{+\infty} dxe^{-x\mathcal{T}} \to \int_{\tau_{uv}^2}^{\tau_{ir}^2} dxe^{-x\mathcal{T}} = \frac{1}{\mathcal{T}}\times e^{-x\mathcal{T}}|_{x=\tau_{ir}^2}^{\tau_{uv}^2}, if\ \mathcal{T}>0, \tag{17}$$

where $e^{-x\mathcal{T}}|_{x=\tau_{ir}^2}^{\tau_{uv}^2} = e^{-\tau_{uv}^2\mathcal{T}} - e^{-\tau_{ir}^2\mathcal{T}}$ and $\tau_{uv} = \frac{1}{\Lambda_{uv}}, \tau_{ir} = \frac{1}{\Lambda_{ir}}$. Here, the ultraviolet cutoff Λ_{uv} plays a role like Λ_{3M}, but much softer, while the infrared cutoff Λ_{ir} is introduced to mimic confinement. For the $\mu \neq 0$ case, $\mathcal{T} = s+\widetilde{\omega}_n'^2+R_B^2 = s+\omega_n^2+R_B^2-\mu'^2+2i\omega_n\mu'$ is a complex function and requires analytic continuation to the complex plane, so we rewrite Equation 17 as

$$\begin{aligned}\frac{1}{\mathcal{T}} &= \begin{cases}\int_0^{+\infty} dxe^{-x\mathcal{T}} \to \int_{\tau_{uv}^2}^{\tau_{ir}^2} dxe^{-x\mathcal{T}} = \frac{1}{\mathcal{T}}\times e^{-x\mathcal{T}}|_{x=\tau_{ir}^2}^{\tau_{uv}^2}, & if\ \Re_{\mathcal{T}}>0\\[2ex] -\int_0^{+\infty} dxe^{x\mathcal{T}} \to -\int_{\tau_{uv}^2}^{\tau_{ir}^2} dxe^{x\mathcal{T}} = \frac{1}{\mathcal{T}}\times e^{x\mathcal{T}}|_{x=\tau_{ir}^2}^{\tau_{uv}^2}, & if\ \Re_{\mathcal{T}}<0\end{cases}\\ &= \frac{1}{\mathcal{T}}\times e^{-Sign(\Re_{\mathcal{T}})x\mathcal{T}}|_{x=\tau_{ir}^2}^{\tau_{uv}^2},\end{aligned} \tag{18}$$

where $\Re_{\mathcal{T}}$ is the real part of $\mathcal{T}$. There also exist some other complex forms of the PT regularization [33], in general, they have this form,

$$\frac{1}{\mathcal{T}} = \frac{\mathcal{T}'}{\mathcal{T}}\times\frac{1}{\mathcal{T}'} = \frac{\mathcal{T}'}{\mathcal{T}}\times\int_0^{+\infty} dxe^{-x\mathcal{T}'} \to \frac{\mathcal{T}'}{\mathcal{T}}\times\int_{\tau_{uv}^2}^{\tau_{ir}^2} d\tau e^{-\tau\mathcal{T}'} = \frac{1}{\mathcal{T}}\times e^{-x\mathcal{T}'}|_{x=\tau_{ir}^2}^{\tau_{uv}^2}, \tag{19}$$

where $\mathcal{T}'$ only needs to satisfy $\Re_{\mathcal{T}'} > 0$. One may also choose to rewrite Equation 19 in many other reasonable forms, such as setting $\mathcal{T}' = |\mathcal{T}|$ or $\mathcal{T}' = i\mathcal{T}$ and so on. For example, we can think $\mathcal{T}' = Sign(\Re_{\mathcal{T}})\mathcal{T}$ in Equation 18. However, we have checked that different forms mainly lead to some numerical differences in the results, instead of obvious qualitative changes. Considering the intrinsic discrepancy of this model with QCD, we choose to use Equation 18 as an example for the following discussions.

3. ~~Zero chemical potential and finite temperature case~~

First, let's begin with considering the $\mu = 0$ and ~~finite~~ T case, which can help us to understand the evolution of the early universe. In this case, many equations can be simplified; for example, $\widetilde{\omega}_n = \omega_n$, $n_+ = n_-$, hence $R_C = 0$.

Turn to R_B, using the 3M regularization Equation 16, Equation 8 becomes

$$R_B = m + \frac{8}{3\pi}\frac{\alpha_{ir}}{m_G^2}\int_0^{\Lambda_{3M}^2}\frac{R_B s^{\frac{1}{2}}}{E_{s,R_B}}\tanh(\frac{E_{s,R_B}}{2T})ds, \tag{20}$$

while using the PT regularization Equation 18, Equation 7 becomes [20,39]

$$R_B = m + \frac{8}{3\sqrt{\pi}}\frac{\alpha_{ir}}{m_G^2}T\int_{\tau_{uv}^2}^{\tau_{ir}^2}\frac{R_B e^{-\tau R_B^2}}{\tau^{\frac{3}{2}}}\theta_2(e^{-4\pi^2T^2\tau})d\tau\ , \tag{21}$$

where $\theta_2(q) = 2q^{\frac{1}{4}}\sum_{n=0}^{+\infty}q^{n(n+1)}$ is the Jacobi theta-function.

4. Zero temperature and finite chemical potential case

Compact stars with their small ratio of temperature to chemical potential, $T/\mu \approx 0$, are well approximated by $T \to 0$ and finite μ. In this case, the Matsubara frequencies become continuous, i.e., $\omega_n \to p_0$, and the sum then becomes an integral over the real axis,

$$\lim_{T\to 0} T\sum_{n=-\infty}^{\infty} f(\omega_n) = \frac{1}{2\pi}\int_{-\infty}^{\infty} dp_0 f(p_0), \tag{22}$$

so

$$\begin{aligned} R_C =& \frac{4}{3\pi}\frac{\alpha_{ir}}{m_G^2}\int_0^{\max\{0,\mu'^2-R_B^2\}} s^{\frac{1}{2}}ds \\ =& \begin{cases} 0, & \mu' < R_B \\ \frac{8}{9\pi}\frac{\alpha_{ir}}{m_G^2}(\mu'^2 - R_B^2)^{\frac{3}{2}}, & \mu' > R_B \end{cases} \end{aligned} \tag{23}$$

which gives the total quark number density as

$$\langle q^\dagger q\rangle = \begin{cases} 0, & \mu' < R_B \\ N_C\frac{(\mu'^2-R_B^2)^{\frac{3}{2}}}{3\pi^2}, & \mu' > R_B \end{cases} \tag{24}$$

consistent with Ref. [61].

Turn to R_B, using Equation 9 and the 3M regularization Equation 16, Equation 8 becomes

$$R_B = m + \frac{8}{3\pi}\frac{\alpha_{ir}}{m_G^2}\int_{\max\{0,\mu'^2-R_B^2\}}^{\Lambda_{3M}^2}\frac{R_B s^{\frac{1}{2}}}{\sqrt{s+R_B^2}}ds, \tag{25}$$

while using the PT regularization Equation 18, Equation 7 becomes

$$R_B = m + \frac{8}{3\pi^2}\frac{\alpha_{ir}}{m_G^2}\int_0^\infty ds\int_{-\infty}^{\infty}dp_0\frac{R_B s^{\frac{1}{2}}}{s+p_0'^2+R_B^2}e^{-Sign(s+R_B^2+p_0^2-\mu'^2)x[s+p_0'^2+R_B^2]}\Big|_{x=\tau_{ir}^2}^{\tau_{uv}^2}, \tag{26}$$

where $p_0' = p_0 + i\mu'$.

5. Results

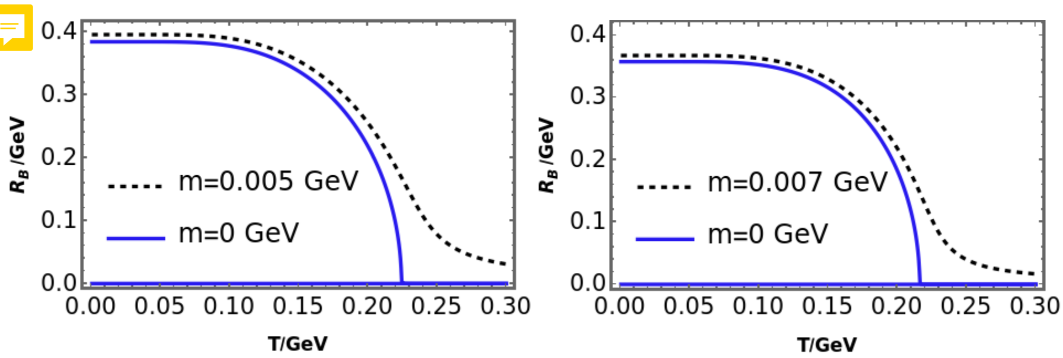
We solve the gap equations using the parameters in ~~Table 1.~~

To match the NJL benchmark, we set $\alpha_{ir} = 0.802\pi$ for the 3M regularization used in Ref.[21] with $N_C = 3, N_f = 2$ so that $G_{CI} = G_{NJL} \approx 5.5GeV^{-2}$ and then slightly retune Λ_{3M} from 0.631 GeV [21] to 0.65 GeV to compensate for the coefficient mismatch between Equation 13 and Equation 15, thereby obtaining results analogous to those of the NJL framework.~~For the 3M regularization we set $\alpha_{ir} = 0.802\pi$ to match the NJL benchmark used in Ref.[21] with $N_C = 3, N_f = 2$ so that $G_{CI} = G_{NJL} \approx 5.5GeV^{-2}$. We then slightly retune Λ_{3M} from 0.631 GeV [21] to 0.65 GeV, so that the vacuum dressed quark mass coincides with the NJL reference.~~[xpli] PT parameters are fixed analogously to reproduce the correct~~same~~[xpli] vacuum solution, ensuring that in-medium differences reflect the impact of different regularization schemes rather than parameter tuning in vacuum.

Solutions of the gap equation with different regularization schemes are shown in Figure 1 and Figure 2, respectively.

Table 1. Parameters used in this work.

Regularization schemes	m **(GeV)**	α_{ir}	m_G **(GeV)**	$\Lambda_{3M}/\Lambda_{uv}$ **(GeV)**	Λ_{ir} **(GeV)**
3M	0.005	0.802π	0.8	0.650	-
PT	0.007	0.93π	0.8	0.905	0.24



(**a**) (**b**)

Figure 1. Solutions of the gap equation with different regularization schemes at $\mu = 0$.(**a**) 3M; (**b**) PT.

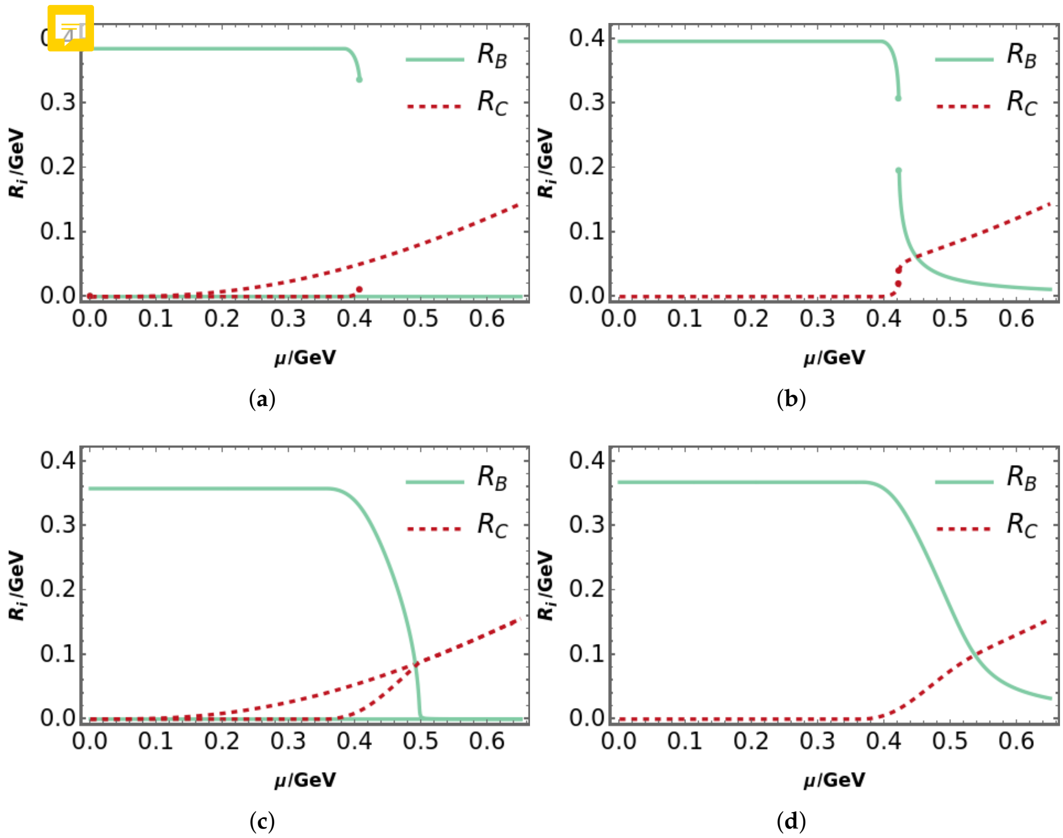


Figure 2. Solutions of the gap equation with different regularization schemes at $T \to 0$. (**a**) 3M in the chiral limit $m = 0$ GeV; (**b**) 3M with $m = 0.005$ GeV; (**c**) PT in the chiral limit; (**d**) PT with $m = 0.007$ GeV.

We can see clearly from Figure 1 that, in the chiral limit $m = 0$ GeV case, the gap equation always has a solution with restored chiral symmetry, i.e. $R_B = 0$, which is usually named the Wigner solution [23,27]. Another solution is the chiral symmetry broken Nambu solution, i.e. $R_B \neq 0$. In domains where the Nambu and Wigner solutions coexist, the thermodynamically stable branch is the one that minimizes the thermodynamic potential. At low temperature this is the Nambu solution. Increasing the temperature, the system undergoes a second-order phase transition to the Wigner phase [27]. However, when beyond the chiral limit, there is only the Nambu solution, and the transition becomes a crossover, which has been discussed widely in the literature. Moreover, it is interesting to note that 3M and PT regularization schemes give very similar results, both qualitatively and quantitatively.

Next, we consider the $T \to 0$ case, as shown in Figure 2. For the PT regularization scheme, (**c**) and (**d**) of Figure 2, the pattern is almost the same as that in Figure 1, besides R_C becoming nonzero and increasing gradually with μ. Namely, in this scheme, the μ effect is very similar to the T one. However, in the 3M regularization, the picture changes a lot, as shown in (**a**) and (**b**) of Figure 2, we can clearly see that there is a first-order phase transition. When beyond the chiral limit, the Nambu and Wigner solutions coexist in the region from $\mu \approx 421.5$ MeV to $\mu \approx 422.0$ MeV. From Figure 2, we also notice that $R_C = 0$ is a good approximation at low μ, which is a popular treatment in the literature; while when μ is large, R_C effect is not negligible.

Comparing Figures 1 and 2, we find that the qualitative effect of increasing T at zero μ is similar for both schemes, whereas with increasing μ at low T the regularization scheme determines the order of the phase transition.

6. Discussion

To clearly show the influence of the vector mean field R_C, we use the 3M parameters from Table 1 and set $\alpha_{ir} = 0.73\pi$. As shown in Figure 3, this turns the previously observed first order phase transition from Figure 2(**b**) into a crossover. If we set $R_C = 0$, the curve will move to lower μ and turn first-order again. Obviously, this shows that R_C has a significant impact on the order of the phase transition and weakens the first order phase transition.

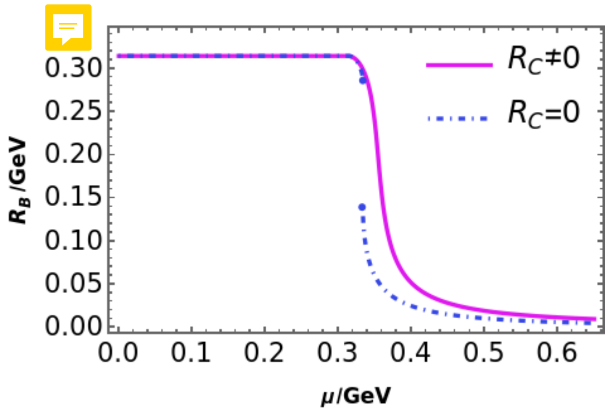


Figure 3. R_C effect on the phase transition in the 3M regularization scheme.

Next, we analyze how the regularization schemes change the order of the phase transition. As mentioned above, since both ~~of them~~ are non-renormalizable, one has to introduce some special treatment to cure the ultraviolet divergence and make the quantities finite. Comparing Equation 25 and Equation 26, we find that the 3M regularization scheme imposes a hard cutoff Λ_{3M}^2 which eliminates contributions from higher momenta. Contrarily, the momentum integration goes up to $+\infty$ in the PT regularization, with a soft cutoff due to the exponential damping factor $e^{-\tau^2\mathcal{T}}$.

To interpolate between hard cutoff and the soft exponential damping factor, we introduce the following equation,

$$R_B = m + \frac{8}{3\pi^2}\frac{\alpha_{ir}}{m_G^2}\int_0^{\Lambda_H^2} ds \int_{-\infty}^{\infty} dp_0 \frac{R_B s^{\frac{1}{2}}}{s + p_0'^2 + R_B^2} e^{-Sign(s+R_B^2+p_0^2-\mu'^2)x[s+p_0'^2+R_B^2]}\Big|_x^{\tau_{uv}^2}, \qquad (27)$$

where $e^{-x\mathcal{T}}\big|_x^{\tau_{uv}^2} = e^{-\tau_{uv}^2\mathcal{T}}$. The exponential factor related to τ_{ir} is approximately zero; we ignore it for this discussion. Using the PT parameterization from Table 1 we vary Λ_{uv} as shown in Table 2. In order to keep the vacuum mass $R_B(T = \mu = 0)$ constant for all Λ_{uv} we adjust Λ_H accordingly. In this way we interpolate from a PT type solution (A) to a 3M type solution (F). Figure 4 shows how at $T = 0$ the PT crossover in $R_B(\mu)$ then gradually changes into a first order phase transition.

Table 2. 6 sets of Λ_{uv} and Λ_H for Equation 27.

Cases	Λ_{uv} (GeV)	Λ_H (GeV)
A	0.905	10
B	1.5	0.786
C	3	0.67
D	5	0.64
E	10	0.62
F	100	0.605

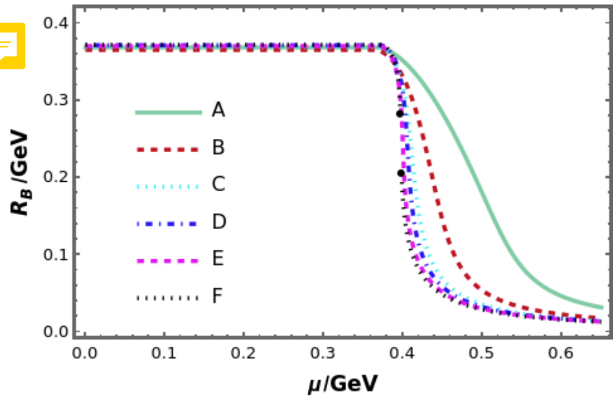


Figure 4. Solutions of Equation 27 using parameters listed in Table 1 and Table 2.

We conclude that, as a noncovariant regularization scheme, the 3M cutoff treatment introduces a hard and sharp cutoff in momentum space, which tends to result in more abrupt changes in the behaviors of the physical quantities (we use effective mass as an example, and it's easy to understand that related quantities have similar patterns), expressed as a first-order phase transition. While the PT regularization is a covariant scheme that smoothly regularizes the interaction by introducing an exponential damping factor, which is gentle and soft, so that most related quantities and processes will also tend to change continuously, expressed as a crossover.

Finally, we discuss the influence of the remaining parameters on the phase transition as listed in Table 3. The results are shown in Figure 5. With PT regularization any combination of Λ_{uv}, α_{ir} and m results in a crossover phase transition at $T \to 0$; For the 3M regularization, we show numerically that the dimensionless quantity $\frac{\alpha_{ir}\Lambda_{3M}^2}{m_G^2}$ determines the order of the phase transition. For example, at $m = 0.005$ GeV, we find numerically that smaller values of $\frac{\alpha_{ir}\Lambda_{3M}^2}{m_G^2} \lesssim 1.6$ result in a crossover, values $\gtrsim 1.6$ ~~in~~ a first-order transition.

[TK 1] If you have a figure to illustrate it that would be fine. Although it's not strictly necessary I think.

Table 3. 4 parameter sets used in Figure 5.

Cases	α_{ir}	Λ_{uv} (GeV)	m (GeV)
A	0.93π	0.905	0.007
B	1.15π	0.905	0.007
C	0.93π	1	0.007
D	0.93π	0.905	0.02

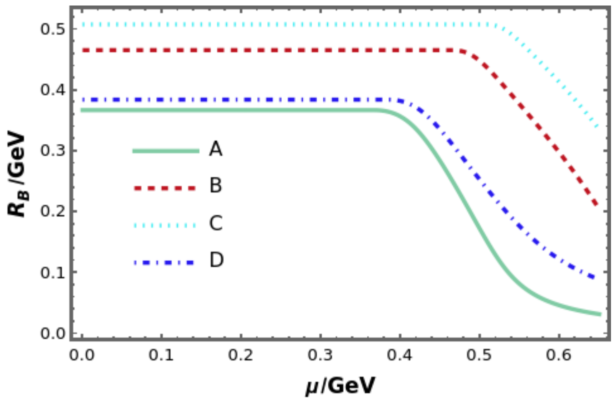


Figure 5. Solutions of Equation 26 using parameters listed in Table 3.

Conclusions

We used a contact interaction to compare the effect of different regularization schemes: the non-covariant three-momentum cutoff and the covariant proper time treatment, both at finite temperature T and chemical potential μ. At zero μ and finite T, the chiral phase transition is a smooth crossover for both regularization schemes. When turning to the low T and finite μ region, different regularization schemes and parameterizations may provide qualitatively different, namely, crossover or first-order. We find that different parameters have only quantitative influence for the covariant regularization, but determine the order of the phase transition for the non-covariant regularization.

[TK 2] Suggest a broader perspective to give a referee something beyond a parameter study: "Our results highlight that for non-renormalizable models (here due to the described properties of the contact-interaction), the chosen renormalization scheme can affect the thermodynamic behavior of a system qualitatively; an uncertainty that should be kept in mind before relying on a model for phenomenological predictions."

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Abbreviations

The following abbreviations are used in this manuscript:

QCD	Quantum chromodynamics
EoS	Equation of state
NJL	Nambu-Jona-Lasinio
DSEs	Dyson-Schwinger equations
3M	Three-momentum
PT	Proper time

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